

PRODUCT DESCRIPTION

Name of Pharmaceutical Product

HP-Daptomycin Powder for solution for infusion 500mg

Pharmacological Therapeutic Class

Daptomycin for injection contains daptomycin, a cyclic lipopeptide antibacterial agent.

Composition

Active Ingredient: Daptomycin

Inactive Ingredient: Sodium hydroxide, Water for injection (WFI) and Nitrogen

Description

Pale yellow to light brown, lyophilized cake or powder.

Reconstituted solution of daptomycin for injection is pale yellow clear solution.

Diluted solution of daptomycin for injection is pale yellow clear solution.

Dosage Form

Type

Lyophilized powder for reconstitution.

Method/Route of Administration

Daptomycin for injection is given by intravenous (IV) infusion.

Sterility Statement

Daptomycin for injection is a sterile product contained in a single-dose vial

INDICATIONS AND USAGE

Daptomycin for injection is indicated for the treatment of the infections listed below.

Complicated Skin and Skin Structure Infections

Adult (≥ 18 years of age) and pediatric (1 to 17 years of age) patients with complicated skin and skin structure infections (cSSSI) caused by susceptible isolates of the following Gram-positive bacteria: *Staphylococcus aureus* (including methicillin-resistant isolates), *Streptococcus pyogenes*, *Streptococcus agalactiae*, *Streptococcus dysgalactiae* subsp. *equisimilis*, and *Enterococcus faecalis* (vancomycin-susceptible isolates only).

Staphylococcus aureus Bloodstream Infections (Bacteremia)

Adult patients (≥ 18 years of age) with *Staphylococcus aureus* bloodstream infections (bacteremia), including those with right-sided infective endocarditis, caused by methicillin-

susceptible and methicillin-resistant isolates.

Pediatric patients (1 to 17 years of age) with *S. aureus* bloodstream infections (bacteremia) caused by methicillin-susceptible and methicillin-resistant isolates.

Limitations of Use

Daptomycin for injection is not indicated for the treatment of pneumonia.

Daptomycin for injection is not indicated for the treatment of left-sided infective endocarditis due to *S. aureus*. The clinical trial of daptomycin for injection in patients with *S. aureus* bloodstream infections included limited data from patients with left-sided infective endocarditis; outcomes in these patients were poor. Daptomycin for injection has not been studied in patients with prosthetic valve endocarditis.

Usage

Appropriate specimens for microbiological examination should be obtained in order to isolate and identify the causative pathogens and to determine their susceptibility to daptomycin.

To reduce the development of drug-resistant bacteria and maintain the effectiveness of daptomycin for injection and other antibacterial drugs, daptomycin for injection should be used only to treat infections that are proven or strongly suspected to be caused by susceptible bacteria.

When culture and susceptibility information is available, it should be considered in selecting or modifying antibacterial therapy. In the absence of such data, local epidemiology and susceptibility patterns may contribute to the empiric selection of therapy. Empiric therapy may be initiated while awaiting test results.

DOSAGE AND ADMINISTRATION

Administration

Daptomycin for injection should be administered intravenously.

Adults

Complicated Skin and Skin Structure Infections

Daptomycin for injection 4 mg/kg should be administered to adult patients intravenously in 0.9% sodium chloride for injection once every 24 hours for 7 to 14 days by infusion over a 30-minute period.

***Staphylococcus aureus* Bloodstream Infections (Bacteremia)**

Daptomycin for injection 6 mg/kg should be administered to adult patients intravenously in 0.9% sodium chloride for injection once every 24 hours for 2 to 6 weeks by infusion over a 30-minute

period. There are limited safety data for the use of daptomycin for injection for more than 28 days of therapy. In the Phase 3 trial, there were a total of 14 patients who were treated with daptomycin for injection for more than 28 days.

Pediatric Patients (1 to 17 Years of Age)

Complicated Skin and Skin Structure Infections

The recommended dosage regimens based on age for pediatric patients with cSSSI are shown in Table 1. Daptomycin for injection should be administered intravenously in 0.9% sodium chloride for injection once every 24 hours for up to 14 days.

Table 1: Recommended Dosage of Daptomycin for injection in Pediatric Patients (1 to 17 Years of Age) with Complicated Skin and Skin Structure Infections, Based on Age

Age group	Dosage*	Duration of therapy
12 to 17 years	5 mg/kg once every 24 hours infused over 30 minutes	Up to 14 days
7 to 11 years	7 mg/kg once every 24 hours infused over 30 minutes	
2 to 6 years	9 mg/kg once every 24 hours infused over 60 minutes	
1 to < 2 years	10 mg/kg once every 24 hours infused over 60 minutes	

*Recommended dosage is for pediatric patients (1 to 17 years of age) with normal renal function. Dosage adjustment for pediatric patients with renal impairment has not been established.

***Staphylococcus aureus* Bloodstream Infections (Bacteremia)**

The recommended dosage regimens based on age for pediatric patients with *S. aureus* bloodstream infections (bacteremia) are shown in Table 2. Daptomycin for injection should be administered intravenously in 0.9% sodium chloride for injection once every 24 hours for up to 42 days.

Table 2: Recommended Dosage of Daptomycin for injection in Pediatric Patients (1 to 17 Years of Age) with *S. aureus* Bloodstream Infections, Based on Age

Age group	Dosage*	Duration of therapy ⁽¹⁾
12 to 17 years	7 mg/kg once every 24 hours infused over 30 minutes	Up to 42 days
7 to 11 years	9 mg/kg once every 24 hours infused over 30 minutes	
1 to 6 years	12 mg/kg once every 24 hours infused over 60 minutes	

*Recommended dosage is for pediatric patients (1 to 17 years of age) with normal renal function. Dosage adjustment for pediatric patients with renal impairment has not been established.

(1) Minimum duration for pediatric bacteremia should be in accordance with the perceived risk of complications in the individual patient

Patients with Renal Impairment

The recommended dosage regimen for adult patients with creatinine clearance (CL_{CR}) <30 mL/min, including those on hemodialysis or continuous ambulatory peritoneal dialysis (CAPD), is 4 mg/kg (cSSSI) or 6 mg/kg (*S. aureus* bloodstream infections) once every 48 hours (Table 3). When possible, daptomycin for injection should be administered following the completion of hemodialysis on hemodialysis days (see Warnings and Precautions, Use in Specific Populations, and Clinical Pharmacology).

The dosage regimen for daptomycin for injection pediatric patients with renal impairment has not been established.

Table 3. Recommended Dosage of Daptomycin for injection in Adult Patients

Creatinine Clearance (CL_{CR})	Dosage Regimen	
	cSSSI	<i>S. aureus</i> Bloodstream Infections
≥30 mL/min	4 mg/kg once every 24 hours	6 mg/kg once every 24 hours
<30 mL/min, including hemodialysis and CAPD	4 mg/kg once every 48 hours*	6 mg/kg once every 48 hours*

*When possible, administer daptomycin for injection following the completion of hemodialysis on hemodialysis days.

Preparation of daptomycin for injection for Administration

Daptomycin for injection is supplied in single-dose vials, each containing 500 mg daptomycin as a sterile, lyophilized powder. The contents of a daptomycin for injection vial should be reconstituted, using aseptic technique, to 50 mg/mL as follows:

Note: To minimize foaming, AVOID vigorous agitation or shaking of the vial during or after reconstitution.

1. Remove the polypropylene flip-off cap from daptomycin for injection vial to expose the central portion of the rubber stopper.
2. Wipe the top of the rubber stopper with an alcohol swab or other antiseptic solution and allow to dry. After cleaning, do not touch the rubber stopper or allow it to touch any other surface.
3. Slowly transfer 10 mL of 0.9% sodium chloride for injection through the center of the rubber stopper into the daptomycin for injection vial, pointing the transfer needle toward the wall of the vial. It is recommended that a beveled sterile transfer needle that is 21 gauge or smaller in diameter, or a needleless device is used, pointing the transfer needle toward the wall of the vial.
4. Ensure that all of the daptomycin for injection powder is wetted by gently rotating the vial.
5. Allow the wetted product to stand undisturbed for 10 minutes.
6. Gently rotate or swirl the vial contents for a few minutes, as needed, to obtain a completely reconstituted solution.
7. Slowly remove reconstituted liquid (50 mg daptomycin/mL) from the vial using a beveled sterile needle that is 21 gauge or smaller in diameter.

Adults

Intravenous Infusion over a period of 30 minutes

- For IV infusion over a period of 30 minutes in adult patients, the appropriate volume of the reconstituted daptomycin for injection (concentration of 50 mg/mL) should be further diluted, using aseptic technique, into a 50 mL IV infusion bag containing 0.9% sodium chloride for injection.

Pediatric Patients (1 to 17 Years of Age)

Intravenous Infusion over a period of 30 or 60 minutes

- For IV infusion over a period of 30 minutes in pediatric patients, reconstituted daptomycin for injection (concentration of 50 mg/mL) is further diluted, using aseptic technique, into a 50 mL IV infusion bag containing 0.9% sodium chloride for injection.

The infusion rate should be maintained at 1.67 mL/min over the 30 minute period.

- For IV infusion over a period of 60 minutes in pediatric patients, reconstituted daptomycin for injection (concentration of 50 mg/mL) is further diluted, using aseptic technique, into an IV infusion bag containing 25 mL of 0.9% sodium chloride for injection.

The infusion rate should be maintained at 0.42 mL/min over the 60-minute period.

Parenteral drug products should be inspected visually for particulate matter prior to administration.

No preservative or bacteriostatic agent is present in this product. Aseptic technique must be used in the preparation of final IV solution. Stability studies have shown that the reconstituted solution is stable in the vial for 12 hours at room temperature and up to 48 hours if stored under refrigeration at 2 to 8°C (36 to 46°F).

The diluted solution is stable in the infusion bag for 12 hours at room temperature and 48 hours if stored under refrigeration at 2 to 8°C (36 to 46°F). The combined storage time (reconstituted solution in vial and diluted solution in infusion bag) should not exceed 12 hours at room temperature or 48 hours under refrigeration at 2 to 8°C (36 to 46°F).

Daptomycin for injection vials are for single use only.

Compatible Intravenous Solutions

The compatible diluent for reconstitution is 0.9% sodium chloride for injection and lactated Ringer's injection.

For IV infusion, reconstituted daptomycin for injection (concentration of 50 mg/mL) is further diluted, using aseptic technique, with 0.9% sodium chloride for injection.

Reconstitute daptomycin for injection, as directed above, to a concentration of 50 mg/mL with 0.9% sodium chloride for injection. Further dilute using aseptic technique with additional 0.9% sodium chloride for injection to a final concentration in the range of 2.5 to 20 mg/mL (typically 10 mg/mL).

Vial Size	Nominal Concentration of Reconstituted Solution	Approximate Available Volume of Reconstituted Solution	Volume of Additional Diluent	Total Volume of Solution for Infusion	Nominal Concentration of Solution for Infusion
500mg	50 mg/mL	10mL	15mL	25 mL	20 mg/mL
500mg	50 mg/mL	10mL	40mL	50 mL	10 mg/mL
500mg	50 mg/mL	10mL	190mL	200 mL	2.5 mg/mL

Incompatibilities

Daptomycin for injection is not compatible with dextrose-containing diluents.

Daptomycin for injection should not be used in conjunction with ReadyMED® elastomeric infusion pumps (Cardinal Health, Inc.). Stability studies of daptomycin for injection solutions stored in ReadyMED® elastomeric infusion pumps identified an impurity (2-mercaptobenzothiazole) leaching from this pump system into the daptomycin for injection solution.

Because only limited data are available on the compatibility of daptomycin for injection with other IV substances, additives and other medications should not be added to daptomycin for injection single-dose vials or infusion bags, or infused simultaneously through the same IV line. If the same IV line is used for sequential infusion of different drugs, the line should be flushed with a compatible intravenous solution before and after infusion with daptomycin for injection.

CONTRAINDICATIONS

Daptomycin for injection is contraindicated in patients with known hypersensitivity to daptomycin or to any of the excipients.

WARNINGS AND PRECAUTIONS

Anaphylaxis/Hypersensitivity Reactions

Anaphylaxis/hypersensitivity reactions have been reported with the use of antibacterial agents, including daptomycin for injection. If an allergic reaction to daptomycin for injection occurs, discontinue the drug and institute appropriate therapy.

Pneumonia

Daptomycin for injection should not be used for the treatment of pneumonia. It has been demonstrated in clinical studies that daptomycin for injection is not effective in the treatment of

community-acquired pneumonia, due to binding to pulmonary surfactant and consequent inactivation.

Skeletal Muscle Effects

Increases in plasma CPK levels, muscular pains, weakness, and/or rhabdomyolysis have been reported during therapy with daptomycin for injection.

It is recommended that:

- Patients receiving daptomycin for injection be monitored for the development of muscle pain or weakness, particularly of the distal extremities.
- In patients who receive daptomycin for injection, CPK levels be measured at baseline and at regular intervals (at least weekly), and more frequently in patients who received recent prior or concomitant therapy with an HMG-CoA reductase inhibitor or in whom elevations in CPK occur during treatment with daptomycin for injection.
- In adult patients with renal impairment, both renal function and CPK be monitored more frequently than once weekly.
- Daptomycin for injection be discontinued in patients with unexplained signs and symptoms of myopathy in conjunction with CPK elevations to levels greater than 1000 U/L (approximately 5 times upper limit of normal [ULN]) and in patients without reported symptoms who have marked elevations in CPK, with levels greater than 2000 U/L ($\geq 10 \times$ ULN).
- Consideration be given to suspending agents associated with rhabdomyolysis, such as HMG- CoA reductase inhibitors, temporarily in patients receiving daptomycin for injection.

Peripheral Neuropathy

Physicians should be alert to signs and symptoms of peripheral neuropathy in patients receiving daptomycin for injection.

Pediatric population

Pediatric patients younger than one year old should not be given daptomycin for injection due to the risk of potential effects on muscular, neuromuscular, and/or nervous systems (either peripheral and/or central) that were observed in neonatal dogs.

Eosinophilic Pneumonia

Eosinophilic pneumonia has been reported in patients receiving daptomycin for injection. In reported cases associated with daptomycin for injection, patients developed fever, dyspnea with hypoxic respiratory insufficiency, and diffuse pulmonary infiltrates or organizing

pneumonia. In general, patients developed eosinophilic pneumonia 2 to 4 weeks after starting daptomycin for injection and improved when daptomycin for injection was discontinued and steroid therapy was initiated. Recurrence of eosinophilic pneumonia upon re-exposure has been reported. Patients who develop these signs and symptoms while receiving daptomycin for injection should undergo prompt medical evaluation, and daptomycin for injection should be discontinued immediately. Treatment with systemic steroids is recommended.

Drug Reaction with Eosinophilia and Systemic Symptoms (DRESS)

DRESS has been reported in post-marketing experience with daptomycin. Patients who develop fever, skin rash, peripheral eosinophilia and systemic organ (for example, hepatic, pulmonary or renal) impairment while receiving daptomycin for injection should undergo medical evaluation. If DRESS is suspected, daptomycin for injection should be discontinued promptly and appropriate treatment instituted.

Tubulointerstitial Nephritis (TIN)

TIN has been reported in post-marketing experience with daptomycin. Patients who develop new or worsening renal impairment while receiving daptomycin for injection should undergo medical evaluation. If TIN is suspected, daptomycin for injection should be discontinued promptly and appropriate treatment instituted.

***Clostridioides difficile*–Associated Diarrhea**

Clostridioides difficile–associated diarrhea (CDAD) has been reported with the use of nearly all antibacterial agents, including daptomycin for injection. If CDAD is suspected or confirmed, daptomycin for injection may need to be discontinued and appropriate treatment instituted as clinically indicated.

Persisting or Relapsing *S. aureus* Bacteremia/Endocarditis

Patients with persisting or relapsing *S. aureus* bacteremia/endocarditis or poor clinical response should have repeat blood cultures. If a blood culture is positive for *S. aureus*, minimum inhibitory concentration (MIC) susceptibility testing of the isolate should be performed using a standardized procedure, and diagnostic evaluation of the patient should be performed to rule out sequestered foci of infection. Appropriate surgical intervention (e.g., debridement, removal of prosthetic devices, valve replacement surgery) and/or consideration of a change in antibacterial regimen may be required.

Drug-Laboratory Test Interactions

False prolongation of prothrombin time (PT) and elevation of International Normalized Ratio (INR) have been observed when certain recombinant thromboplastin reagents are utilized for the assay (see Undesirable Effects, Drug-Laboratory Test Interactions).

Non-Susceptible Microorganisms

The use of antibacterials may promote the overgrowth of non-susceptible microorganisms. If superinfection occurs during therapy, take appropriate measures.

UNDESIRABLE EFFECTS

Undesirable Effects Identified during Clinical Trials

During clinical trials of daptomycin for injection, the following adverse drug reactions were reported during therapy and during follow-up. The adverse drug reactions are organized by system organ class, and the frequency categories for these adverse drug reactions are reported in the table below as follows:

Very common:	$\geq 1/10$ ($\geq 10\%$)
Common:	$\geq 1/100$ and $< 1/10$ ($\geq 1\%$ and $< 10\%$)
Uncommon:	$\geq 1/1000$ and $< 1/100$ ($\geq 0.1\%$ and $< 1\%$)
Rare:	$\geq 1/10,000$ and $< 1/1000$ ($\geq 0.01\%$ and $< 0.1\%$)
Very rare:	$< 1/10,000$ ($< 0.01\%$)

Adverse Drug Reaction	Frequency Category
Blood and lymphatic system disorders	
Anemia	Common
Thrombocytosis	Uncommon
Eosinophilia	Uncommon
Leukocytosis	Uncommon
Cardiac disorders	
Supraventricular arrhythmia	Uncommon
Ear and labyrinth disorders	
Vertigo	Uncommon
Gastrointestinal disorders	
Gastrointestinal and abdominal pain	Common
Diarrhea	Common
Vomiting	Common

Flatulence, bloating, and distension	Common
Constipation	Common
Nausea	Common
Dyspepsia	Uncommon
Abdominal distension	Uncommon
General disorders and administration site conditions	
Asthenia	Common
Pyrexia	Common
Infusion site reaction	Common
Chills	Uncommon
Fatigue	Uncommon
Hepatobiliary disorders	
Jaundice	Rare
Infections and infestations	
Urinary tract infection	Common
Fungal infection	Common
Candida infection	Common
Fungemia	Uncommon
Investigations	
Blood creatine phosphokinase increased	Common
Liver function test abnormal (increased ALT, AST, or ALP)	Common
Blood creatinine increased	Uncommon
International Normalized Ratio increased	Uncommon
Blood lactate dehydrogenase increased	Uncommon
Prothrombin time prolonged	Rare
Metabolism and nutrition disorders	
Hyperglycemia	Uncommon
Electrolyte imbalance	Uncommon
Decreased appetite	Uncommon
Musculoskeletal, connective tissue, and bone disorders	
Limb pain	Common
Muscular weakness	Uncommon
Muscle pain	Uncommon
Arthralgia	Uncommon
Muscle cramps	Uncommon

Nervous system disorders	
Dizziness	Common
Headache	Common
Paresthesia	Uncommon
Tremor	Uncommon
Taste disorder	Uncommon
Eye irritation	Uncommon
Psychiatric disorders	
Anxiety	Common
Insomnia	Common
Renal and urinary disorders	
Renal impairment, including renal failure and renal insufficiency	Uncommon
Reproductive system and breast disorders	
Vaginitis	Uncommon
Skin and subcutaneous tissue disorders	
Pruritus	Common
Rash	Common
Urticaria	Uncommon
Vascular disorders	
Hypertension	Common
Hypotension	Common
Flushing	Uncommon

Undesirable Effects Reported Post-Marketing

The following adverse drug reactions, not listed above, have been reported during worldwide post- marketing experience:

Blood and lymphatic system disorders

Thrombocytopenia

Immune system disorders

Hypersensitivity reactions, including, but not limited to, anaphylaxis, angioedema and pulmonary eosinophilia

Infections and infestations

Clostridioides difficile–associated diarrhea

Investigations

Myoglobin increased, platelet count decreased

Musculoskeletal, connective tissue, and bone disorders

Rhabdomyolysis

Nervous system disorders

Peripheral neuropathy

Renal and urinary disorders

Tubulointerstitial nephritis (TIN)

Respiratory, thoracic, and mediastinal disorders

Cough

Eosinophilic pneumonia

Organizing pneumonia

Skin and subcutaneous tissue disorders

Vesiculobullous rash with or without mucous membrane involvement (Stevens-Johnson Syndrome (SJS) or Toxic Epidermal Necrolysis (TEN))

Drug reaction with eosinophilia and systemic symptoms (DRESS)

Acute generalized exanthematous pustulosis

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES

No studies on the effects on the ability to drive and use machines have been performed.

On the basis of reported adverse drug reactions, daptomycin is presumed to be unlikely to produce an effect on the ability to drive or use machinery.

INTERACTIONS**Interactions with Other Medicinal Products**

Daptomycin for injection was studied in adult human drug-drug interaction studies with aztreonam, tobramycin, warfarin, simvastatin, and probenecid. Daptomycin had no effect on the pharmacokinetics of warfarin or probenecid, nor did these drugs alter the pharmacokinetics of daptomycin. The pharmacokinetics of daptomycin were not significantly altered by aztreonam.

Experience with the concomitant administration of daptomycin for injection and warfarin is limited. Studies of daptomycin for injection with anticoagulants other than warfarin have not been conducted. Monitor anticoagulant activity in patients receiving daptomycin for injection and warfarin for the first several days after therapy with daptomycin for injection is initiated.

Experience with the coadministration of HMG-CoA reductase inhibitors and daptomycin for injection in patients is limited; therefore, consider suspending use of HMG-CoA reductase inhibitors temporarily in patients receiving daptomycin for injection.

Although small changes in the pharmacokinetics of daptomycin and tobramycin were observed during coadministration by IV infusion over a 30-minute period using a daptomycin for injection dose of 2 mg/kg, the changes were not statistically significant. The interaction between daptomycin and tobramycin with a clinical dose of daptomycin for injection is unknown. Caution is warranted when daptomycin for injection is coadministered with tobramycin.

Drug-Laboratory Test Interactions

Clinically relevant plasma concentrations of daptomycin have been observed to cause a significant concentration-dependent false prolongation of prothrombin time (PT) and elevation of International Normalized Ratio (INR) when certain recombinant thromboplastin reagents are utilized for the assay. The possibility of an erroneously elevated PT/INR result due to interaction with a recombinant thromboplastin reagent may be minimized by drawing specimens for PT or INR testing near the time of trough plasma concentrations of daptomycin. However, sufficient daptomycin concentrations may be present at trough to cause interaction.

If confronted with an abnormally high PT/INR result in a patient being treated with daptomycin for injection, it is recommended that clinicians:

1. Repeat the assessment of PT/INR, requesting that the specimen be drawn just prior to the next daptomycin for injection dose (i.e., at trough concentration). If the PT/INR value obtained at trough remains substantially elevated above what would otherwise be expected, consider evaluating PT/INR utilizing an alternative method.
2. Evaluate for other causes of abnormally elevated PT/INR results.

USE IN SPECIFIC POPULATIONS

Pregnancy and Lactation

Risk Summary

Embryo/fetal development studies performed in rats and rabbits at doses of up to 75 mg/kg (approximately 2 and 4 times the recommended 6 mg/kg human dose, respectively, on a body

surface area basis) revealed no evidence of harm to the fetus due to daptomycin. Daptomycin can cross the placenta in pregnant rats. There are, however, no adequate and well-controlled studies in pregnant women. Because animal reproduction studies are not always predictive of human response, daptomycin for injection should be used during pregnancy only if the potential benefit outweighs the possible risk.

Excretion of daptomycin into milk of lactating animals has not been studied. In a single human case study, daptomycin for injection was administered daily for 28 days to a nursing mother at an IV dose of 6.7 mg/kg/day, and samples of the patient's breast milk were collected over a 24-hour period on day 27. The highest measured concentration of daptomycin in the breast milk was 0.045 µg/mL, which is a low concentration. Until more experience is gained, women should be instructed to avoid breast-feeding while receiving daptomycin for injection.

Pediatric

The safety and effectiveness of daptomycin for injection in patients 1 to 17 years are supported by evidence from adequate and well-controlled studies in adults, pharmacokinetic data in pediatric patients, and additional data from two prospective studies in pediatric patients 1 to 17 years of age with cSSSI and pediatric patients 2 to 17 years of age with *Staphylococcus aureus* Bloodstream Infections (bacteremia).

In clinical trials, 372 pediatric patients (3 months to 17 years of age) were given intravenous daptomycin for injection. Pharmacokinetic studies enrolled a total of 61 pediatric patients, and an additional 256 and 55 pediatric patients received daptomycin for injection in the prospective studies of cSSSI (DAP-PEDS-07-03) and bacteremia (DAP-PEDBAC-11-02), respectively.

Elderly

No adjustment of daptomycin for injection dosage is warranted for elderly patients with $CL_{CR} \geq 30$ mL/min.

Renal Impairment

Daptomycin is eliminated primarily by the kidneys; therefore, an adjustment of daptomycin for injection dosage interval is recommended for adult patients with $CL_{CR} < 30$ mL/min, including patients receiving hemodialysis or CAPD. The recommended dosing regimen for these adult patients is 4 mg/kg (cSSSI) or 6 mg/kg (*S. aureus* bloodstream infections) once every 48 hours. Alternatively, adult patients on hemodialysis can be dosed three times per week. When possible, administer daptomycin for injection following the completion of hemodialysis on hemodialysis days. In adult patients with renal impairment, monitor both renal function and CPK more frequently than once weekly.

No dosage interval adjustment is required for adult patients with $CL_{CR} \geq 30$ mL/min.

The dosage regimen for daptomycin for injection in pediatric patients with renal impairment has not been established.

Hepatic Impairment

No dosage adjustment is warranted when daptomycin for injection is administered to patients with mild to moderate hepatic impairment (Child-Pugh Class B). The pharmacokinetics of daptomycin in patients with severe hepatic impairment (Child-Pugh Class C) have not been evaluated.

Gender

No dosage adjustment is warranted based on gender when daptomycin for injection is administered.

Obesity

No adjustment of daptomycin for injection dosage is warranted in obese patients.

OVERDOSE

In the event of overdose, supportive care is advised. Daptomycin is cleared slowly from the body by hemodialysis (approximately 15% of the administered dose is removed over 4 hours) and by peritoneal dialysis (approximately 11% of the administered dose is removed over 48 hours).

CLINICAL PHARMACOLOGY

Pharmacokinetic Properties

General Characteristics

Daptomycin pharmacokinetics were generally linear (dose-proportional) and time-independent at daptomycin for injection doses of 4 to 12 mg/kg administered by IV infusion over a 30-minute period as a single daily dose for up to 14 days in adults. Steady-state concentrations were achieved by the third daily dose.

Distribution

Daptomycin is reversibly bound to human plasma proteins (mean binding range of 90 to 93%) in a concentration-independent manner, and serum protein binding trended lower (mean binding range of 84 to 88%) in adult subjects with significant renal impairment ($CL_{CR} < 30$ mL/min or on dialysis). The protein binding of daptomycin in adult subjects with mild to

moderate hepatic impairment (Child-Pugh Class B) was similar to that in healthy adult subjects.

The volume of distribution at steady-state of daptomycin in healthy adult subjects was approximately 0.1 L/kg and was independent of dose. Tissue distribution studies in rats showed that daptomycin appears to penetrate the blood-brain barrier and the placental barrier only minimally following single and multiple doses.

Metabolism

In *in vitro* studies, daptomycin was not metabolized by human liver microsomes. *In vitro* studies with human hepatocytes indicate that daptomycin does not inhibit or induce the activities of the following human cytochrome P450 isoforms: 1A2, 2A6, 2C9, 2C19, 2D6, 2E1, and 3A4. It is unlikely that daptomycin will inhibit or induce the metabolism of drugs metabolized by the P450 system.

After infusion of ^{14}C -daptomycin in healthy adults, the plasma radioactivity was similar to the concentration determined by microbiological assay. Inactive metabolites were detected in urine, as determined by the difference between total radioactive concentrations and microbiologically active concentrations. In a separate study, no metabolites were observed in plasma, and minor amounts of three oxidative metabolites and one unidentified compound were detected in urine. The site of metabolism has not been identified.

Elimination

Daptomycin is excreted primarily by the kidneys. There is minimal to no active tubular secretion of daptomycin. In a mass balance study of adult subjects using radiolabeled daptomycin, 78% of the administered dose was recovered from the urine based on total radioactivity, while urinary recovery of unchanged daptomycin was approximately 52% of the dose. About 6% of the administered dose was excreted in the feces based on total radioactivity. Plasma clearance of daptomycin is approximately 7 to 9 mL/h/kg, and its renal clearance is 4 to 7 mL/h/kg.

Specific Populations

Elderly

The pharmacokinetics of daptomycin were evaluated in 12 healthy elderly subjects (≥ 75 years of age) and 11 healthy young adult controls (18 to 30 years of age). Following administration of a single 4 mg/kg dose of daptomycin for injection by IV infusion over a 30-minute period, the mean total clearance of daptomycin was approximately 35% lower and the mean AUC was

approximately 58% higher in elderly subjects than in healthy young adult subjects. There were no differences in $C_{\max}$.

Renal Impairment

Following administration of a single 4 mg/kg or 6 mg/kg dose of daptomycin for injection by IV infusion over a 30-minute period to adult subjects with various degrees of renal impairment, total daptomycin clearance was lower and systemic exposure (AUC) was higher than in subjects with normal renal function. The mean AUC for patients with $CL_{CR} < 30$ mL/min and for patients on dialysis (CAPD and hemodialysis dosed post-dialysis) was approximately 2 and 3 times higher, respectively, than for patients with normal renal function.

Hepatic Impairment

The pharmacokinetics of daptomycin were evaluated in 10 adult subjects with moderate hepatic impairment (Child-Pugh Class B) and compared with those in healthy adult volunteers (N=9) matched for gender, age, and weight. The pharmacokinetics of daptomycin were not altered in subjects with moderate hepatic impairment. The pharmacokinetics of daptomycin in patients with severe hepatic impairment (Child-Pugh Class C) have not been evaluated.

Pediatric

The pharmacokinetics of daptomycin in pediatric subjects was evaluated in 3 single-dose pharmacokinetic studies. In general, body weight-normalized total body clearance in pediatric patients was higher than in adults and increased with a decrease of age, whereas elimination half-life tends to decrease with a decrease of age. Body weight-normalized total body clearance and elimination half-life of daptomycin in children 2 to 6 years of age were similar at different doses.

A study was conducted to assess safety, efficacy, and pharmacokinetics of daptomycin in pediatric patients (1 to 17 years old, inclusive) with cSSSI caused by Gram-positive pathogens. Patients were enrolled into 4 age groups, and intravenous daptomycin for injection doses of 5 to 10 mg/kg once daily were administered. Following administration of multiple doses, daptomycin exposure (AUC_{ss} and $C_{\max,ss}$) was similar across different age groups after dose adjustment based on body weight and age (Table 4).

Table 4: Mean (SD) Daptomycin Population Pharmacokinetic Parameters in cSSSI Pediatric Patients

Age	Pharmacokinetic Parameters					
	Dose (mg/kg)	AUC _{ss} (mcg•h/mL)	t _{1/2} (h)	V _{ss} (mL)	CL _T (mL/h/kg)	C _{max,ss} (mcg/mL)
12 to 17 years (N=6)	5	434 (67.9)	7.1 (0.9)	8200 (3250)	11.8 (2.15)	76.4 (6.75)
7 to 11 years (N=2)	7	543*	6.8*	4470*	13.2*	92.4*
2 to 6 years (N=7)	9	452 (93.1)	4.6 (0.8)	2750 (832)	20.8 (4.29)	90.3 (14.0)
1 to <2 years (N=27)	10	462 (138)	4.8 (0.6)	1670 (446)	23.1 (5.43)	81.6 (20.7)

AUC_{ss}, area under the concentration-time curve at steady state; CL_T, clearance normalized to body weight;

V_{ss}, volume of distribution at steady state; t_{1/2}, terminal half-life

*Mean is calculated from N=2

A study was conducted to assess safety, efficacy, and pharmacokinetics of daptomycin in pediatric patients (1 to 17 years old, inclusive) with SAB. Patients were enrolled into 3 age groups and intravenous doses of 7 to 12 mg/kg once daily were administered. Following administration of multiple doses, daptomycin exposure (AUC_{ss} and C_{max,ss}) was similar across different age groups after dose adjustment based on body weight and age (Table 5).

Table 5: Mean (SD) of Daptomycin Population Pharmacokinetic Parameters in Bacteremia Pediatric Patients

Age	Pharmacokinetic Parameters						
	Dose (mg/kg)	Infusion Duration (min)	AUC _{ss} (mcg•h/mL)	T _½ (hr)	V _{ss} (mL)	CL _T (mL/h/kg)	C _{max,ss} (mcg/mL)
12 to 17 years (N=13)	7	30	656 (334)	7.5 (2.3)	6420 (1980)	12.4 (3.9)	104 (35.5)
7 to 11 years (N=19)	9	30	579 (116)	6.0 (0.8)	4510 (1470)	15.9 (2.8)	104 (14.5)
2 to 6 years (N=19)	12	60	620 (109)	5.1 (0.6)	2200 (570)	19.9 (3.4)	106 (12.8)

AUC_{ss}, area under the concentration-time curve at steady state; CL_T, clearance normalized to body weight; V_{ss}, volume of distribution at steady state; t_½, terminal half-life

No patients 1 to <2 years of age were enrolled in the study. Simulation using a population pharmacokinetic model demonstrated that the AUC_{ss} of daptomycin in pediatric patients 1 to <2 years of age receiving 12 mg/kg once daily would be comparable to that in adult patients receiving 6 mg/kg once daily.

Gender

No clinically significant gender-related differences in daptomycin pharmacokinetics have been observed.

Obesity

The pharmacokinetics of daptomycin were evaluated in 6 moderately obese (Body Mass Index [BMI] 25 to 39.9 kg/m²) and 6 extremely obese (BMI ≥40 kg/m²) adult subjects. The AUC was approximately 30% higher in moderately obese subjects and 31% higher in extremely obese subjects than in nonobese controls.

Microbiology

Daptomycin belongs to the cyclic lipopeptide class of antibacterials. Daptomycin is a natural product that has clinical utility in the treatment of infections caused by aerobic, Gram-

positive bacteria. The *in vitro* spectrum of activity of daptomycin encompasses most clinically relevant Gram-positive pathogenic bacteria. Daptomycin retains potency against Gram-positive bacteria that are resistant to other antibacterials, including isolates resistant to methicillin, vancomycin, and linezolid.

Mechanism of Action

The mechanism of action of daptomycin is distinct from that of any other antibacterial. Daptomycin binds to bacterial cell membranes and causes a rapid depolarization of membrane potential. This loss of membrane potential causes inhibition of DNA, RNA, and protein synthesis, which results in bacterial cell death.

Mechanism of Resistance

The mechanism(s) of daptomycin resistance is not fully understood. There are no known transferable elements that confer resistance to daptomycin.

There is no cross-resistance due to resistance mechanisms that are specific for another class of antibacterials.

Emergent decreases in susceptibility have been observed in both *S. aureus* and *enterococcal* isolates following daptomycin for injection therapy.

PK/PD Relationship

Daptomycin exhibits rapid, concentration-dependent bactericidal activity against Gram-positive bacteria *in vitro* and in *in vivo* animal models.

Interactions with Other Antibacterials

In vitro studies have investigated daptomycin interactions with other antibacterials. Antagonism, as determined by kill curve studies, has not been observed. *In vitro* synergistic interactions of daptomycin with aminoglycosides, β -lactam antibacterials, and rifampin have been shown against some isolates of *staphylococci* (including some methicillin-resistant isolates) and *enterococci* (including some vancomycin-resistant isolates).

HOW SUPPLIED

Available Commercial Size

Daptomycin for injection is supplied in single-dose vials containing 500 mg daptomycin as a sterile, lyophilized powder.

Nature and Contents of Container

Single-dose 20 mL vial, containing a pale yellow to light brown, lyophilized cake or powder.

Storage

Store original packages at refrigerated temperatures, 2 to 8°C (36 to 46°F); avoid excessive heat.

Shelf Life

After reconstitution: Chemical and physical stability of the reconstituted solution in the vial has been demonstrated for 12 hours at 25°C and up to 48 hours if stored under refrigeration (2 to 8°C). Chemical and physical stability of the diluted solution in infusion bags has been established as 12 hours at 25°C and 48 hours if stored under refrigeration at 2 to 8°C. The combined storage time (reconstituted solution in vial and diluted solution in infusion bag) must not exceed 12 hours at 25°C or 48 hours at 2 to 8°C.

MANUFACTURER

Hainan Poly Pharm. Co., Ltd.
Guilinyang Economic Development Area,
Haikou, Hainan Province,
571127, China

DATE OF REVISION

16/08/2024